Time Series Analysis of Fertilizer & Food Commodity Prices

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1. Executive Summary

Fertilizer is one of the most important commodities in global food production, and periodic price shocks, such as the 2008 spike, 2022 shock following the Ukraine War, and recent tensions around the Strait of Hormuz, raises the question of how much the shocks affect the food commodities consumers end up paying for. This report has three primary research questions: 1. Which food prices move most closely with fertilizer prices? 2. Does knowing the fertilizer price actually improve forecasts of food prices relative to a model that uses only the food price's own history? 3. Is any effect delayed in a way that is consistent with the growing season?

The monthly data from the World Bank Pink sheet was used, and the dates ranged from January 1960 through February 2026. Five fertilizer price series (DAP, TSP, Urea, KCI, and Phosphate Rock) were combined into a single equal-weighted index. This index was screened against roughly 60 commodity series using lagged cross-correlations, then filtered to food commodities within a 3-12 month response window. US HRW Wheat came out as the most responsive food commodity, with a peak correlation at a nine-month lag.

Then two competing forecasting models were fit for Wheat. A standalone MA(1) & GARCH(1,1) model used only Wheat's own price history. A time-series regression of wheat on the nine-month lagged fertilizer index, with ARIMA(2,1,3) errors, incorporated fertilizer as a predictor. Both are forecast forward to Feburary 2028.

Based on the model diagnostic, the two models are having a similar level of performance. The examination on forecasting ability gaves a similar result. The actual March 2026 Wheat price was $275.91/mt. The GARCH model predicted $293.89/mt and the ARIMA regression model predicted $258.06/mt.

2. Introduction

a. Motivation

Fertilizers are among the most important factors in global crop production, making them the most significant input cost as well. When fertilizer prices undergo extreme price movements, such as during the 2008 financial crisis, the 2022 commodity shock, and Russia's invasion of Ukraine, it raises a natural question: how would the cost increase propagate into food commodity prices? Moreover, the recent events in the Strait of Hormuz impact the fertilizer market too. Understanding this relationship has direct implications for food security policy, commodity market forecasting, and most importantly, our daily lives and cost of living.

b. Research Question

This report will be organized around three research questions. First, which food commodity prices are most strongly associated with fertilizer prices? Second, does incorporating fertilizer price information improve forecasts relative to a standalone ARIMA model? Third, is the effect of fertilizer prices on food prices delayed, consistent with the lag imposed by agricultural growing seasons?

c. Data

The World Bank's Commodity Markets dataset, which contains monthly price observations spanning from January 1960 through February 2026, was used for this project. On the predictor side, we consider five fertilizer price series: Diammonium Phosphate (DAP), Triple Superphosphate (TSP), Urea, Potassium Chloride (KCI), and Phosphate Rock. On the outcome side, the dataset includes a broad range of commodities from grains (maize, wheat, rice, barley, sorghum), oils (soybean, palm, sunflower, coconut), beverages (coffee, tea), and livestock (beef, chicken). The report will initiate from filtering through the commodities to choose which commodities to focus on. All prices are reported in nominal US dollars per metric ton.

3. Methods

a. Exploratory Data Analysis

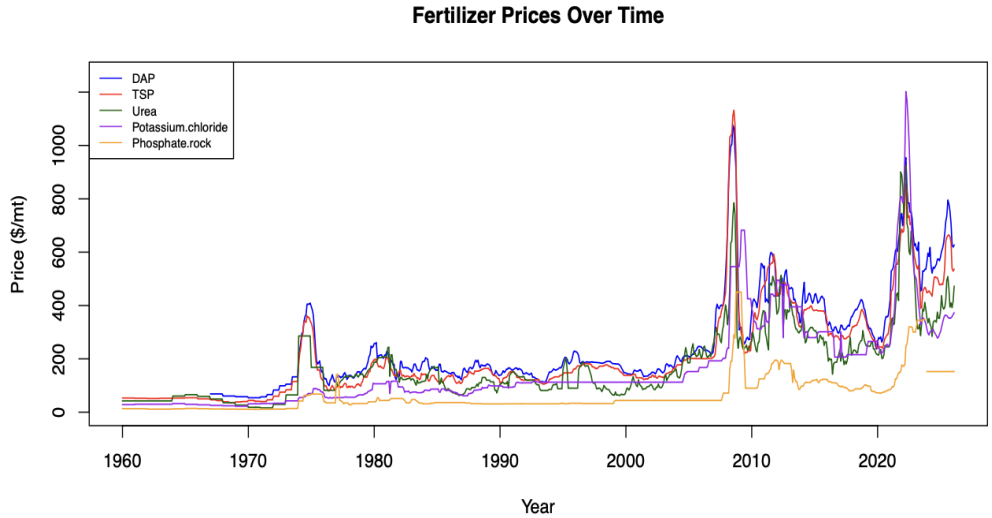

Figure 1: 5 Fertilizers Prices Over Time

The above figure plots the five fertilizer prices from 1960 to 2026. A couple features clearly stand out. All five series share pronounced spikes around 2008 and 2022, suggesting they are driven by common market forces. The series show clear non-stationarity from an upward trend over time and heteroskedasticity. The five fertilizers also appear to be highly correlated with each other, which leads to an idea of combining them into a single index for screening.

b. Detrending Methods

In order to remove some of the trend seen in the financial data, we employ two common methods of standardizing long term. The first is to account for inflation, which is especially prevalent when the dataset has decades of prices of various goods.

To account for inflation, we use the following formula:

$$\text{Real Price}_t = \text{Nominal Price}_t \times \left(\frac{\text{CPI}_{base}}{\text{CPI}_t}\right)$$

where base CPI is the most recent month in the dataset so everything is in "current dollars" with respect to the most current data. In order to get the Consumer Price Index from 1960 to 2026, we downloaded the dataset from FRED and used R to transform the data such that it reflects what the equivalent price of each commodity would be in today's money. Note that while we did this for all commodities, we only graph the fertilizers for continuity sake from the previous section's EDA.

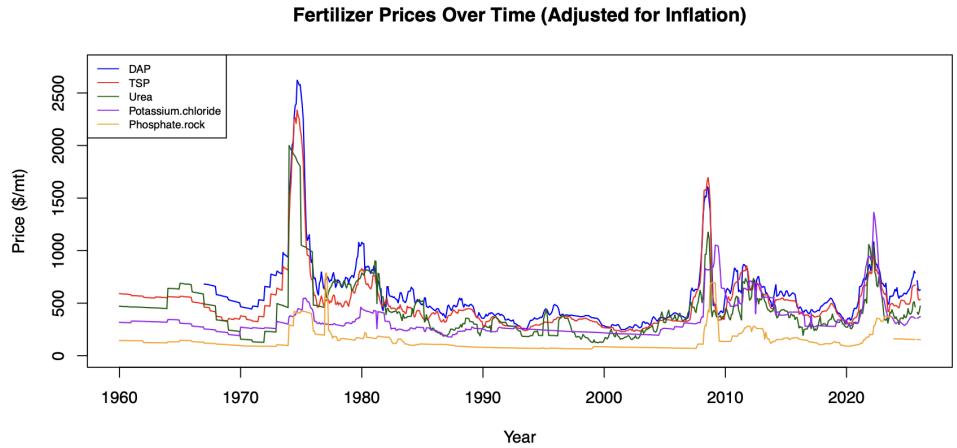

Figure 2: Fertilizer Prices After Adjusting For Inflation Based on CPI

Next, we handle any remaining trending by applying a differenced, logarithmic transformation. Differencing the data will let us evaluate the return on investment while eliminating some of the overall trend. Taking the log will also remove any further exponential growth that is normally expected with financial assets that is especially prevalent in data that spans over multiple decades. After the differenced, log transformation on all commodities, our fertilizer prices look as follows:

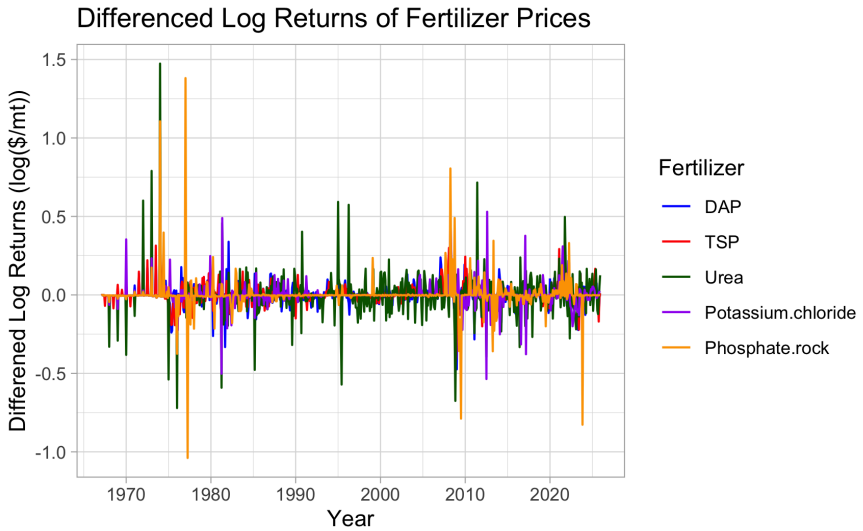

Figure 3: Inflation Adjusted Fertilizer Prices After Taking the Log and Differncing

This looks like a much more stationary dataset that then allows us to further proceed with data analysis.

c. Combining Fertilizer Prices

Fertilizer prices seem highly correlated, and including all fertilizer series as linear predictors would introduce severe multicollinearity. So it makes sense to combine them and compute a single composite score. Therefore, we applied Principal Component Analysis (PCA), commonly used for dimensionality reduction, to the log-differenced real fertilizer prices, which are the monthly inflation-adjusted percentage changes. Log-differenced real fertilizer prices allow us to capture the common short-run movement in fertilizer prices after removing the long-run inflation trend. The results of PCA show that the first principal component (PC1) accounts for about 43.8% of the total shared variance, making it the largest common factor, as shown in Table #1.

PC	variance_explained	cumulative
PC1	**43.8%**	**43.8%**
PC2	20.9%	64.7%
PC3	16.0%	80.6%
PC4	12.9%	93.5%
PC5	6.5%	100.0%

Table 1: Variance Explained and Cumulative Variance by Principal Component

In addition, the correlation between PC1 factor and the composite index with transparent equal weights on each fertilizer is about 0.964, meaning that the PC1 factor and the simple equal-weight average of the log-differenced real fertilizer prices are almost identical. So we calculate the fertilizer composite return index as the equal-weighted index = 0.2*Phosphate.rock + 0.2*DAP + 0.2*TSP + 0.2*Urea + 0.2*KCL for easier interpretation.

d. Screening

The original dataset contains prices for more than 20 commodities. To streamline the analysis, we focused on the commodity with the strongest lagged correlation to fertilizer prices. We measured the lagged correlations between candidate commodity returns and the equal-weighted fertilizer index across positive lags. Each commodity was then ranked by its maximum absolute lagged correlation, along with its best-response lag. The final ranked result is shown in Table 2 after applying economic filters that removed commodities unlikely to be causally linked to fertilizer prices (e.g., crude oil, coal, metals) and those outside economically plausible response windows (3–12 months).

Crop <chr>	Max_Correlation <dbl>	Best_Lag_Months <dbl>
Wheat..US.HRW	0.11199965	9
Soybean.meal	0.10839324	9
Banana..US	0.09557471	4
Tea..avg.3.auctions	0.09313025	3
Sugar..world	0.08946028	10
Sorghum	0.08880164	9
Wheat..US.SRW	0.08434931	9
Fish.meal	0.07380923	10
Nickel	0.07304537	12
Plywood	-0.06404227	8

Table 2: Ranked-Cross-Correlations between Food Prices and Combined Fertilizer Prices and the Corresponding Best Lagged Month

Based on the above ranking, the price for US HRW What ("Wheat") was chosen for the subsequent modeling analysis.

4. Results

a. GARCH models

According to the PCA analysis, the five fertilizer prices were combined into one price index with equal weights (Figure 3). We can see that the price is approximately having a constant mean but with obvious volatility. Therefore, it might be appropriate to fit a GARCH model on the returns.

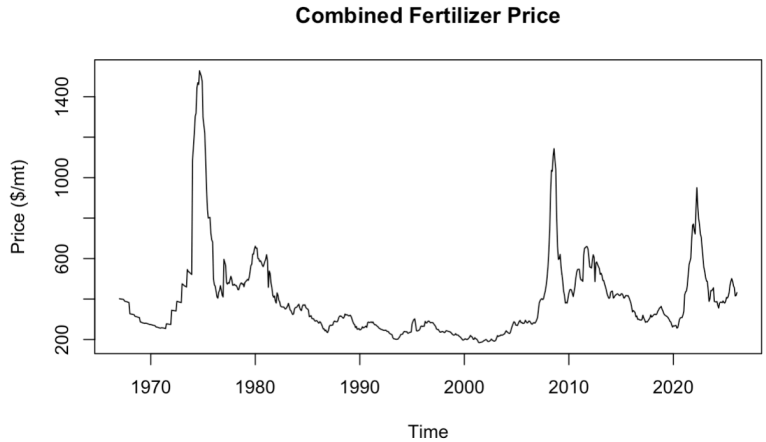

Figure 4: Combined Fertilizer Price

From the ACF and PACF of the squared returns (Figure 4), we observed significant spikes at non-zero lags, which confirmed our use of a GARCH model. Additionally, the ACF and PACF of returns also showed significant spikes, meaning that an ARMA model would be needed for the mean equation. Accordingly, we fit an AR(1)-GARCH(1,1) model for the combined fertilizer price return (Table 3)

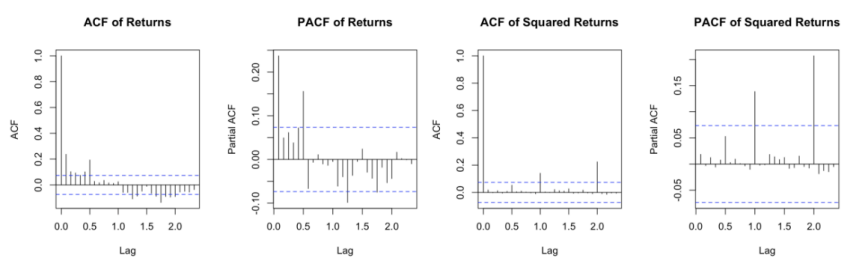

Figure 5: Analysis on Returns and Squared Returns for Combined Fertilizer Price

	Estimate	SD	p-value
mu	-0.001	0.0015	0.502
ar1	0.37	0.047	**< 0.001** ***
omega	4.69E-05	1.15E-05	**< 0.001** ***
alpha1	0.16	0.023	**< 0.001** ***
beta1	0.88	0.014	**< 0.001** ***

Table 3: Summary Output for the AR(1)-GARCH(1,1) Model on Combined Food Price Returns

Similar to the combined fertilizer price, it would be reasonable to fit a GARCH model for the Wheat price that has noticeable heteroskedasticity (Figure 6).

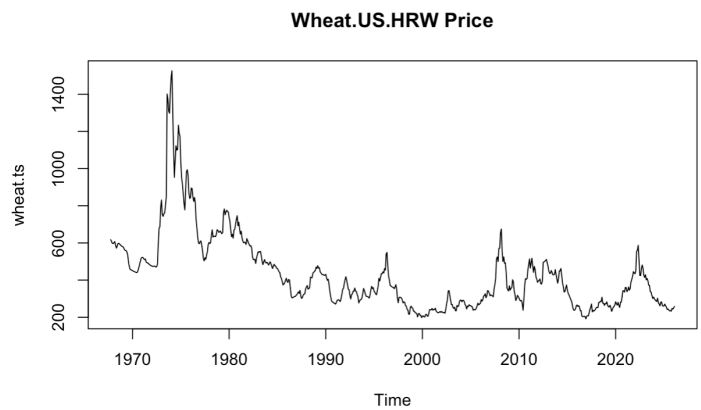

Figure 6: Wheat Price

Again, the use of a GARCH model was further supported by the significant spikes in ACF and PACF for the squared returns. The ACF and PACF of returns both showed a significant spike at lag 1, so it would be reasonable to fit an AR(1) or MA(1) model for the mean equation. A MA(1)-GARCH(1,1) model was chosen for the final modeling of Wheat price (Table 4).

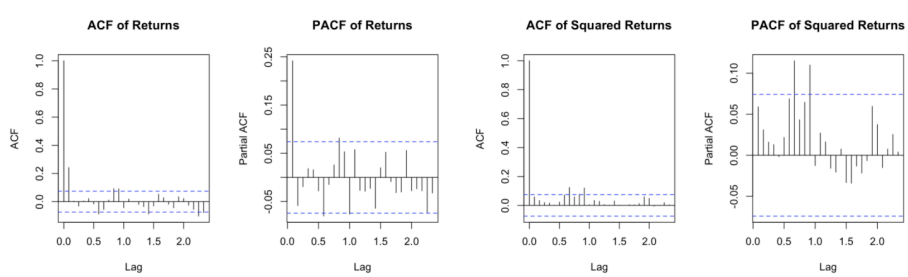

Figure 6: Analysis on Returns and Squared Returns for Wheat Price

	Estimate	SD	p-value
mu	-3.30E-03	2.39E-03	0.167
ar1	0.24	0.039	**< 0.001** ***
omega	1.35E-04	3.75E-05	**< 0.001** ***
alpha1	0.096	0.021	**< 0.001** ***
beta1	0.87	0.022	**< 0.001** ***

Table 4: Summary Output for the MA(1)-GARCH(1,1) Model on Combined Food Price Returns

b. OLS Regression with White Noise Residuals

Based on the screening results, we identified a peak cross-correlation between wheat price and the combined fertilizer price with a 9-months lag (Table 2, Figure 7). Accordingly, we were motivated to fit a regression model of Wheat price on the lagged combined fertilizer price.

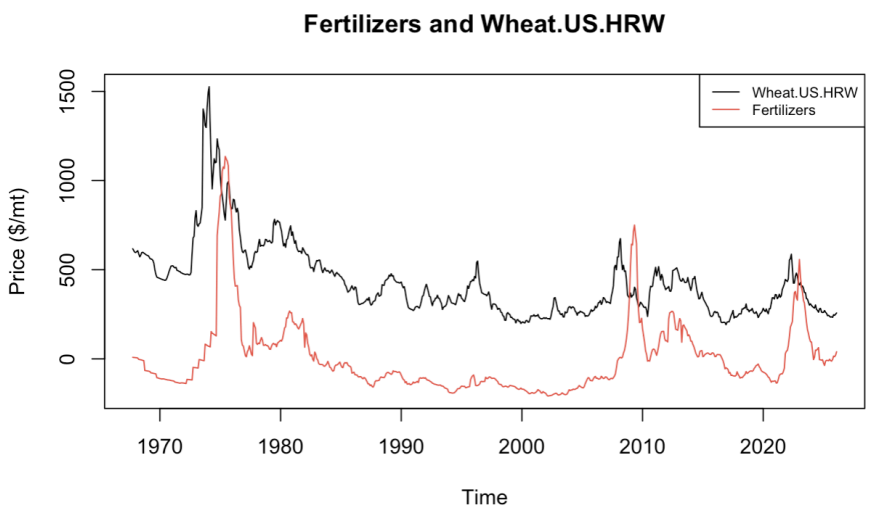

Figure 7: Wheat Prices and Nine-Month Lagged Combined Fertilizer Price

We first fitted an OLS regression model with white noise residuals:

$$Wheat_t = \beta_0 + \beta_1 \times \overline{Fertlizer_{t-9}} + x_t,$$
$$x_t \sim WN(0, \sigma^2)$$

The coefficients are reported in Table 5.

	Estimate	SD	p-value
Intercept	430.41	6.60	**< 0.001** ***
Fertilier	0.52	0.032	**< 0.001** ***

Table 5: Summary Output from OLS Regression of Wheat Price on Mean Lagged Combined Fertilizer Price with White Noise Residuals

We then examined the overall pattern, ACF, and PACF of residuals from the actual fitted model (Figure 8). The plot of residuals showed a clear downward trend. The Q-Q norm plot also showed significant deviation from the diagonal line for tail values. And most importantly, significant spikes were observed in the ACF and PACF plots, suggesting that the residuals were clearly not white noise. As a result, we decided to refit the model with ARIMA(p, 1, q) residuals.

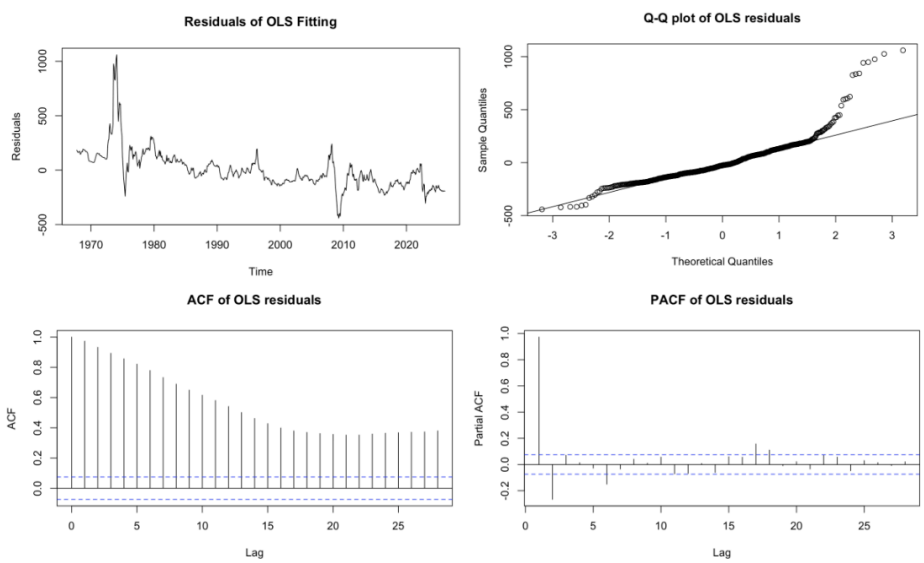

Figure 8: Residual Analysis for OLS Model with White Noise Residuals

c. Regression with ARIMA Residuals

Using the auto. arima function (specifying seasonality and stationarity to false), we refit a model with ARIMA(2,1,3) residuals. The coefficient on the mean lagged combined fertilizer price is 0.084 (Table 4), suggesting that a one-unit increase in the USD price above the mean is associated with an estimated 0.084 USD increase in the Wheat price. This effect size is notably smaller than that for the white noise OLS fit.

	Estimate	SD
ar1	0.44	0.051
ar2	-0.91	0.038
ma1	-0.19	0.066
ma2	0.81	0.056
ma3	0.15	0.043
xreg	0.084	0.037

Table 4: Summary Output from OLS Regression of Wheat Price on Mean Lagged Combined Fertilizer Price with ARIMA(2,1,3) Residuals

We again conducted an analysis on the residuals from this refitted model (Figure 9). The residuals seem to have a mean of zero and are approximately normally distributed with slight deviations from the diagonal line in the Q-Q plot. While some significant spikes persist in the ACF and PACF plots, the residuals for the refitted ARIMA model were more similar to white noise than the previous OLS model.

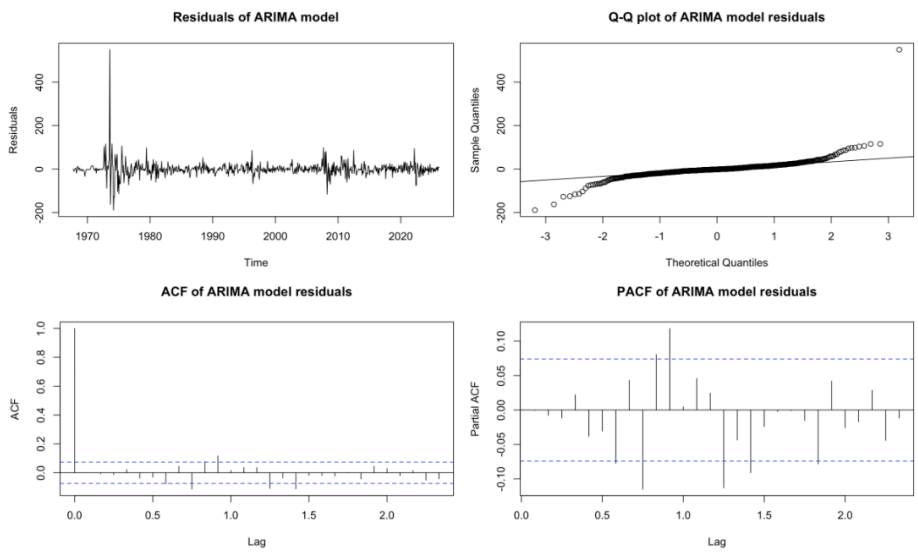

Figure 9: Residual Analysis for the Refitted ARIMA Model

In addition to residuals, we also did comparisons between observed and fitted values for the two models (Figure 10). It could be seen that the fitted Wheat price is noticeably more aligned with the observed value, which further indicates that the ARIMA model is a better model choice.

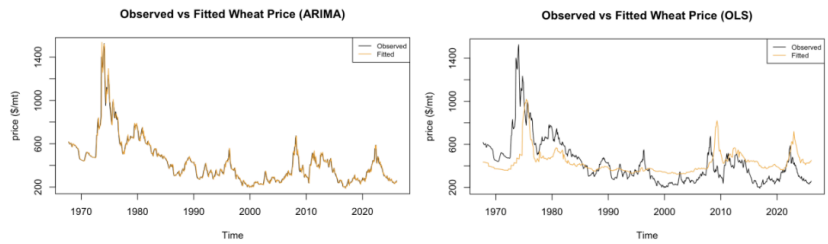

Figure 10: Observed vs Fitted Values

d. Forecasts

With the above analysis, two different models were obtained for the Wheat price: the GARCH model and the ARIMA regression model. To answer the research question of whether knowing fertilizer prices improves predictions on food prices, we decided to examine the forecasting performance of these two models.

For the GARCH model, predictions were first made for the Wheat price returns. The exponential of the cumulative sum of returns was then calculated as the forecasts of Wheat price on the original scale for the next 24 months from 2026M03 to 2028M02 (Figure 11). Because of the nature of the model, the estimations were smoother than the observed values, and the confidence intervals were wider for later times in the future.

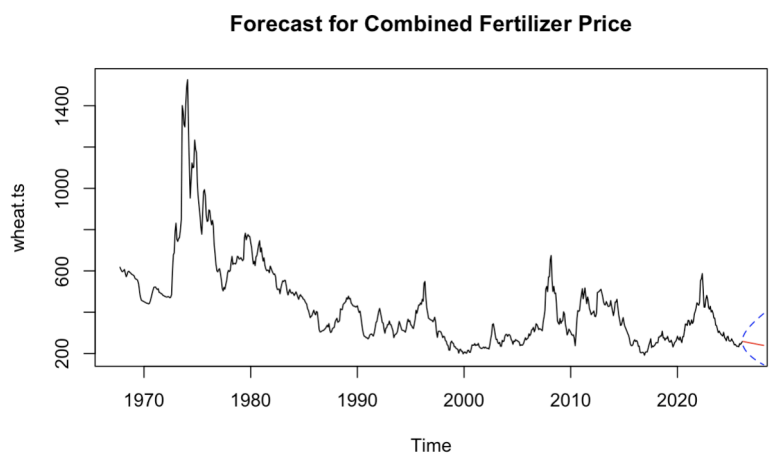

Figure 11: Forecasts (in Red) for theWheat Price

For the regression model, covariate values for future times are required to make predictions. Therefore, we estimated combined fertilizer prices on their original scale based on the GARCH model (Figure 12). The estimated combined fertilizer prices were then input into the ARIMA regression model to make predictions for future Wheat prices (Figure 13).

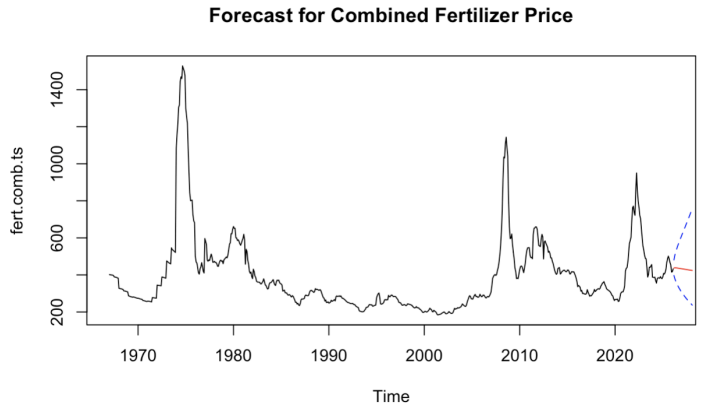

Figure 12: Forecasts (in Red) for the Combined Fertilizer Price

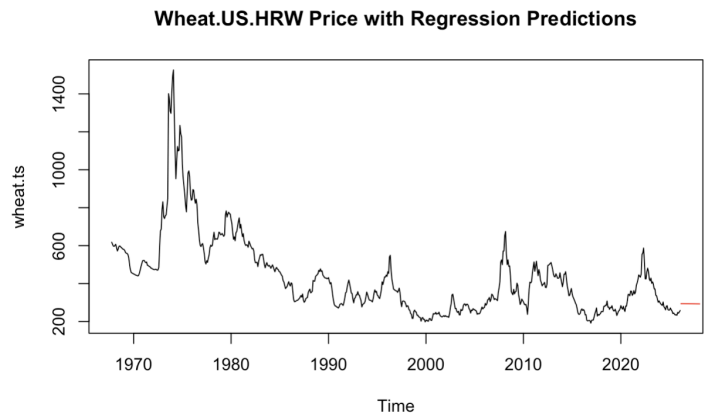

Figure 13: Forecasts for Wheat Price (in Red) with ARIMA Regression Model

By April 2026, the World Bank's Commodity Markets dataset updates the newest commodity prices for 2026M03. Based on that, the real Wheat price in 2026M03 is $275.91/mt. The forecast for the Wheat price in the same month from the GARCH model is $293.89/mt, and that from the ARIMA regression is $258.06/mt. Examining the difference between actual value and predicted value, the two models had a similar performance.

5. Discussion

a. Why Wheat

Wheat was selected for the modeling analysis because it ranked the highest among all candidate commodities after the lagged cross-correlation screening. This ranking indicates that Wheat exhibits the strongest delayed response to short-run movements of fertilizer composite return index, making it the most relevant outcome variable for studying the transmission of fertilizer price shocks.

According to our background review in agriculture, wheat is a fertilizer-intensive cereal crop, and cereal accounts for a large share of global mineral fertilizer use. This is economically relevant because wheat production depends heavily on the three primary plant nutrients, nitrogen (N) , phosphorus (P) , and potassium (K), especially around sowing and early growth, as stated in the FAO wheat description that "for good yields the fertilizer requirements are up to 150 kg/ha N, 35 to 45 kg/ha P and 25 to 50 kg/ha K" (FAO). The fertilizers in our composite index—DAP, TSP, Urea, KCl, and Phosphate Rock—represent the main nitrogen (N), phosphorus (P), and potassium (K) nutrient channels used in crop production. Specifically, urea is a major nitrogen source, DAP provides nitrogen and phosphorus, TSP is a concentrated phosphorus fertilizer, KCl is a major potassium source, and phosphate rock is the mineral feedstock for many phosphorus fertilizers (IFDC).

b. Economic Interpretation of the Lag

The estimated peak lag of approximately 9 months between fertilizer price movements and Wheat price responses is economically meaningful, as it is consistent with agricultural production cycles. Fertilizers are applied during the planting and early growth stages, and their effects on crop yield manifest only after the growing and harvesting processes are completed. According to the FAO's Wheat overview, "the length of the total growing period of winter wheat needs about 180 to 250 days to mature" (FAO) , which is about 6-8 months, making a multi-month delayed response in wheat market prices reasonable. Therefore, the identified lag structure supports the interpretation that fertilizer price shocks affect Wheat prices with a delay rather than instantly.

c. Interpretation of Effect Size

According to our ARIMA regression output, although the estimated coefficient on the lagged fertilizer index is statistically significant, its magnitude is relatively small (0.084). This suggests that fertilizer price movements do have a measurable impact on Wheat prices, but they are not the only determinant of price fluctuations and do not dominate price formation. Other factors, such as weather conditions and geopolitical events, could also play important roles in shaping commodity prices.

Therefore, the relatively small effect size is consistent with the multifactorial nature of agricultural markets.

d. Limitations of the Analysis

Several limitations of our study should be acknowledged. First, the analysis is based on observational time series data, so the identified relationships should be interpreted as associations rather than causal effects. Therefore, even though lagged correlations have some economic plausibility, unobserved confounding factors may still influence both fertilizer and commodity prices. Second, the model does not explicitly account for other important driving factors of agricultural prices, such as weather variability and energy costs, so it may bias the prediction of Wheat prices. Finally, the model assumes a stable lag structure over time, which may not hold during periods of major world changes, such as geopolitical shocks or policy interventions.

6. Conclusion

a. Summary of Findings

In this study, we examined the relationship between fertilizer prices and food commodity prices, particularly using lagged fertilizer prices as a predictor. Using an index constructed from the five fertilizer series, we found that wheat had the strongest relationship. Modeling using GARCH and ARIMA residuals regression, we found that the forecasts are able to capture the general directional movements of wheat prices. However, residual checks on these models suggest that they are bad at predicting outliers due to the noisy nature of financial data.

b. Future Extensions

There are several directions in which this analysis could be extended. As discussed, the model does not take into account data that naturally affects agricultural prices such as climate and policies. Thus, a natural extension is to add in additional covariates such as climate patterns and market events to increase the model complexity and hence accuracy. Moreover, the current models only consider first-degree (linear) relationships between wheat prices and fertilizer prices. It is plausible that higher-degree relationships exist, and we can use feature engineering to construct polynomial terms to run higher-order regressions.

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